

3 Theory of Quotients—First Principles of Theory of Numbers

Exercises III

1. Express in its simplest form—

$$\frac{x^2}{x-y} + \frac{y^2}{y-x}.$$

$$\begin{aligned} \frac{x^2}{x-y} + \frac{y^2}{y-x} &= \frac{x^2(y-x)}{(x-y)(y-x)} + \frac{y^2(x-y)}{(x-y)(y-x)} & \left(\frac{a}{b} + \frac{p}{q} = \frac{qa}{qb} + \frac{pb}{qb}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{(x-y)(y-x)} & \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{xy - x^2 - y^2 + xy} & \text{(expand denominator)} \\ &= \frac{x^2(y-x) + y^2(x-y)}{x(y-x) + y(x-y)} & \text{(factor denominator)} \\ &= \frac{x^2(y-x) - y^2(y-x)}{x(y-x) - y(y-x)} & (x-y = -(y-x)) \\ &= \frac{(x^2 - y^2)(y-x)}{(x-y)(y-x)} & \text{(extract common factor } y-x) \\ &= \frac{(x^2 - y^2)}{(x-y)} & \left(\frac{ac}{bc} = \frac{a}{b}\right) \\ &= \frac{(x-y)(x+y)}{(x-y)} & (a^2 - b^2 = (a-b)(a+b)) \\ &= x+y & \left(\frac{ac}{bc} = \frac{a}{b}\right) \end{aligned}$$

2. Express in its simplest form—

$$\frac{a}{a-b} + \frac{b}{b-a}.$$

$$\begin{aligned} \frac{a}{a-b} + \frac{b}{b-a} &= \frac{a(b-a) + b(a-b)}{(a-b)(b-a)} && \left(\frac{a}{b} + \frac{p}{q} = \frac{aq}{bq} + \frac{bp}{bq}\right) \\ &= \frac{a(b-a) + b(a-b)}{ab - a^2 - b^2 + ab} && \text{(expand denominator)} \\ &= \frac{a(b-a) + b(a-b)}{a(b-a) + b(a-b)} && \text{(factor denominator)} \\ &= 1 && \left(\frac{a}{a} = 1\right) \end{aligned}$$

3. Simplify—

$$\frac{P+Q}{P-Q} - \frac{P-Q}{P+Q}, \quad \text{where } P = x+y, \quad Q = x-y.$$

$$\begin{aligned} \frac{P+Q}{P-Q} - \frac{P-Q}{P+Q} &= \frac{x+y+x-y}{x+y-x+y} - \frac{x+y-x+y}{x+y+x-y} && \text{(replace variables)} \\ &= \frac{2x}{2y} - \frac{2y}{2x} && \text{(simplify)} \\ &= \frac{x}{y} - \frac{y}{x} && \left(\frac{ab}{bc} = \frac{a}{c}\right) \\ &= \frac{x^2 - y^2}{xy} && \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \end{aligned}$$

4. Simplify—

$$\frac{1 - \frac{x(1-y)}{x+y}}{1 + \frac{1-y}{x+y}}.$$

$$\begin{aligned}
1 - \frac{x(1-y)}{x+y} \div 1 + \frac{1-y}{x+y} &= \frac{x+y}{x+y} - \frac{x(1-y)}{x+y} \div \frac{x+y}{x+y} + \frac{1-y}{x+y} \\
&= \frac{x+y-x+xy}{x+y} \div \frac{x+y+1-y}{x+y} \\
&= \frac{xy+y}{x+y} \div \frac{x+1}{x+y} \\
&= \frac{xy+y}{x+y} \times \frac{x+y}{x+1} \\
&= \frac{xy+y}{x+1} \\
&= \frac{y(x+1)}{x+1} \\
&= y
\end{aligned}$$

5. Simplify—

$$\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}}$$

$$\begin{aligned}
\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}} &= \left(\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc} \right) \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} = a \div b \right) \\
&= \left(\frac{c}{abc} - \frac{b}{abc} - \frac{a}{abc} \right) \div \frac{a^2 - (b-c)^2}{a} && \text{(common denominator } abc) \\
&= \frac{c-b-a}{abc} \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b} \right) \\
&= \frac{c-b-a}{abc} \div \frac{(a-(b-c))(a+(b-c))}{a} && (a^2 - b^2 = (a-b)(a+b)) \\
&= \frac{c-b-a}{abc} \div \frac{(a-b+c)(a+b-c)}{a} && \text{(remove brackets)} \\
&= \frac{c-b-a}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p} \right) \\
&= \frac{-(a+b-c)}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && (c-b-a = -(a+b-c)) \\
&= \frac{-a(a+b-c)}{abc(a-b+c)(a+b-c)} && \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= -\frac{1}{bc(a-b+c)} && \left(\frac{ac}{bc} = \frac{a}{b} \right)
\end{aligned}$$

6. Simplify—

$$\left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right).$$

$$\begin{aligned} & \left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right) \\ &= \left(\frac{a(a+b)}{a+b} - \frac{b^2}{a+b}\right) \times \left(\frac{a(a-b)}{a-b} + \frac{b^2}{a-b}\right) \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \left(\frac{a^2+ab-b^2}{a+b}\right) \times \left(\frac{a^2-ab+b^2}{a-b}\right) \quad \left(\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}\right) \\ &= \frac{(a^2+ab-b^2)(a^2-ab+b^2)}{(a+b)(a-b)} \quad \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq}\right) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{(a+b)(a-b)} \quad (\text{expand numerator}) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{a^2 - b^2} \quad (\text{distribute denominator}) \end{aligned}$$

7. Simplify—

$$\frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2}.$$

$$\begin{aligned} & \frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2} \\ &= \frac{x-y+x+y}{(x+y)(x-y)} - \frac{2x}{x^2+y^2} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \frac{2x}{x^2-y^2} - \frac{2x}{x^2+y^2} \quad (\text{collect like terms; } (a+b)(a-b) = a^2 - b^2) \\ &= \frac{2x(x^2+y^2) - 2x(x^2-y^2)}{(x^2-y^2)(x^2+y^2)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq-pb}{bq}\right) \\ &= \frac{2x^3 + 2xy^2 - 2x^3 + 2xy^2}{(x^2-y^2)(x^2+y^2)} \quad (\text{expand numerator}) \\ &= \frac{4xy^2}{x^4 + x^2y^2 - y^2x^2 - y^4} \quad (\text{collect like terms; expand denominator}) \\ &= \frac{4xy^2}{x^4 - y^4} \quad (\text{collect like terms}) \end{aligned}$$

8. Simplify—

$$\left(\frac{x}{y} + \frac{y}{x}\right) \left(\frac{a}{b} + \frac{b}{a}\right) - \left(\frac{x}{y} - \frac{y}{x}\right) \left(\frac{a}{b} - \frac{b}{a}\right).$$

$$\begin{aligned}
& \left(\frac{x}{y} + \frac{y}{x} \right) \left(\frac{a}{b} + \frac{b}{a} \right) - \left(\frac{x}{y} - \frac{y}{x} \right) \left(\frac{a}{b} - \frac{b}{a} \right) \\
&= \frac{x^2 + y^2}{xy} \times \frac{a^2 + b^2}{ab} - \frac{x^2 - y^2}{xy} \times \frac{a^2 - b^2}{ab} & \left(\frac{a}{b} \pm \frac{p}{q} = \frac{aq \pm bp}{bq} \right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2)}{abxy} - \frac{(x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2) - (x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b} \right) \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - (a^2x^2 - b^2x^2 - a^2y^2 + b^2y^2)}{abxy} & \text{(expand numerator)} \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - a^2x^2 + b^2x^2 + a^2y^2 - b^2y^2}{abxy} & \text{(remove brackets)} \\
&= \frac{2b^2x^2 + 2a^2y^2}{abxy} & \text{(collect like terms)} \\
&= \frac{2(b^2x^2 + a^2y^2)}{abxy} & \text{(extract common factor 2)}
\end{aligned}$$

9. Simplify—

$$\left(\frac{a}{b} - \frac{b}{a} \right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2} \right).$$

$$\begin{aligned}
& \left(\frac{a}{b} - \frac{b}{a} \right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2} \right) \\
&= \frac{a^2 - b^2}{ab} \div \frac{a^4 - b^4}{a^2b^2} & \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{a^4 - b^4} & \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p} \right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{(a^2 - b^2)(a^2 + b^2)} & (a^2 - b^2 = (a - b)(a + b)) \\
&= \frac{a^2b^2(a^2 - b^2)}{ab(a^2 - b^2)(a^2 + b^2)} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= \frac{ab}{a^2 + b^2} & \left(\frac{ac}{bc} = \frac{a}{b} \right)
\end{aligned}$$

10. Simplify—

$$\frac{a(a - b) - b(a + b)}{\frac{a}{a + b} - \frac{b}{a - b}}.$$

$$\begin{aligned}
& \frac{a(a-b) - b(a+b)}{\frac{a}{a+b} - \frac{b}{a-b}} \\
&= (a(a-b) - b(a+b)) \div \left(\frac{a}{a+b} - \frac{b}{a-b} \right) \quad \left(\frac{a}{b} = a \div b \right) \\
&= (a(a-b) - b(a+b)) \div \frac{a(a-b) - b(a+b)}{(a+b)(a-b)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= (a(a-b) - b(a+b)) \times \frac{(a+b)(a-b)}{a(a-b) - b(a+b)} \quad \left(a \div \frac{p}{q} = a \times \frac{q}{p} \right) \\
&= (a+b)(a-b) \quad \left(\frac{ac}{bc} = \frac{a}{b} \right) \\
&= a^2 - b^2 \quad ((a+b)(a-b) = a^2 - b^2)
\end{aligned}$$

11. Simplify—

$$\frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2}.$$

$$\begin{aligned}
& \frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2} \\
&= \frac{(1-x)(1-x+x^2) - (1+x)(1+x+x^2)}{(1+x+x^2)(1-x+x^2)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= \frac{1-2x+2x^2-x^3 - (1+2x+2x^2+x^3)}{(1+x+x^2)(1-x+x^2)} \quad (\text{expand numerator}) \\
&= \frac{-4x-2x^3}{(1+x^2)^2 - x^2} \quad ((p+q)(p-q) = p^2 - q^2) \\
&= \frac{-4x-2x^3}{1+x^2+x^4} \quad (\text{expand denominator})
\end{aligned}$$

12. Simplify—

$$\frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1}.$$

$$\begin{aligned}
& \frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1} \\
&= \frac{\frac{1+2x}{1+x}}{\frac{x^2+x+1}{1+x}} \div \frac{2x+1}{x^2+x+1} \quad (\text{common denominator } 1+x; \text{ expand}) \\
&= \frac{1+2x}{1+x} \times \frac{1+x}{x^2+x+1} \div \frac{2x+1}{x^2+x+1} \quad \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p}\right) \\
&= \frac{1+2x}{x^2+x+1} \times \frac{x^2+x+1}{2x+1} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

13. Simplify—

$$\frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3}.$$

$$\begin{aligned}
& \frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3} \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{1}{a} + \frac{1}{b}\right)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}, \text{ cancelling } \frac{1}{a} + \frac{1}{b}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{a+b}{ab}\right)^2} \quad \left(\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{a^2 b^2}{(a+b)^2} \quad \left(\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2ab}{(a+b)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= \frac{a^2 + 2ab + b^2}{(a+b)^2} \quad \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\
&= \frac{(a+b)^2}{(a+b)^2} \quad (a^2 + 2ab + b^2 = (a+b)^2) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

14. Show that—

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)}$$

is independent of x .

Proof. Taking $a^2b^2(a^2 - b^2)$ as a common denominator,

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)} = \frac{N}{a^2b^2(a^2 - b^2)},$$

where

$$\begin{aligned} N &= x^4(a^2 - b^2) + b^2(x^2 - a^2)^2 - a^2(x^2 - b^2)^2 \\ &= x^4(a^2 - b^2) + b^2(x^4 - 2a^2x^2 + a^4) \\ &\quad - a^2(x^4 - 2b^2x^2 + b^4) && ((p - q)^2 = p^2 - 2pq + q^2) \\ &= a^2x^4 - b^2x^4 + b^2x^4 - 2a^2b^2x^2 + a^4b^2 \\ &\quad - a^2x^4 + 2a^2b^2x^2 - a^2b^4 && (\text{remove brackets}) \\ &= a^4b^2 - a^2b^4 && (\text{collect like terms}) \\ &= a^2b^2(a^2 - b^2) && (\text{extract common factor } a^2b^2) \end{aligned}$$

Then the expression is

$$\frac{a^2b^2(a^2 - b^2)}{a^2b^2(a^2 - b^2)} = 1.$$

From this we see that all terms of x cancel, so its value is always 1, thus the expression is independent of x . $\square$

15. Simplify—

$$\frac{a}{b - \frac{c}{d - \frac{e}{f}}}.$$

$$\begin{aligned}
& \frac{a}{b - \frac{c}{d - \frac{e}{f}}} \\
&= a \div \left[b - \left(c \div \left(d - \frac{e}{f} \right) \right) \right] & \left(\frac{a}{b} = a \div b \right) \\
&= a \div \left[b - \left(c \div \frac{df - e}{f} \right) \right] & \left(a - \frac{p}{q} = \frac{aq - p}{q} \right) \\
&= a \div \left[b - \frac{cf}{df - e} \right] & \left(a \div \frac{p}{q} = \frac{aq}{p} \right) \\
&= a \div \left[\frac{b(df - e) - cf}{df - e} \right] & \left(a - \frac{p}{q} = \frac{aq - p}{q} \right) \\
&= \frac{a(df - e)}{b(df - e) - cf} & \left(a \div \frac{p}{q} = \frac{aq}{p} \right) \\
&= \frac{adf - ae}{bdf - be - cf} & (\text{expand})
\end{aligned}$$

16. Simplify—

$$\frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}}$$

Writing $t = a - 2b$,

$$\begin{aligned}
& \frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}} \\
&= \frac{1}{t - \frac{1}{t - \frac{1}{t}}} \quad (\text{replace variables}) \\
&= \frac{1}{t - \frac{1}{\frac{t^2 - 1}{t}}} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}\right) \\
&= \frac{1}{t - \frac{t}{t^2 - 1}} \quad \left(a \div \frac{p}{q} = \frac{aq}{p}\right) \\
&= \frac{1}{\frac{t(t^2 - 1) - t}{t^2 - 1}} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}\right) \\
&= \frac{t^2 - 1}{t^3 - 2t} \quad (\text{expand; } a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{(a - 2b)^2 - 1}{(a - 2b)^3 - 2(a - 2b)} \quad (\text{restore variables})
\end{aligned}$$

17. Simplify—

$$\frac{a + b}{a + b + \frac{1}{a - b + \frac{1}{a + b}}}.$$

$$\begin{aligned}
& \frac{a+b}{a+b+\frac{1}{a-b+\frac{1}{a+b}}} \\
&= \frac{a+b}{a+b+\frac{1}{(a-b)(a+b)+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{a+b+\frac{a+b}{a^2-b^2+1}} \quad ((a+b)(a-b) = a^2-b^2; a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+1) + (a+b)}{a^2-b^2+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+2)}{a^2-b^2+1}} \quad (\text{extract common factor } a+b) \\
&= \frac{(a+b)(a^2-b^2+1)}{(a+b)(a^2-b^2+2)} \quad \left(a \div \frac{p}{q} = \frac{aq}{p}\right) \\
&= \frac{a^2-b^2+1}{a^2-b^2+2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right)
\end{aligned}$$

Exercises IV

1. If the two fractions A/B , a/b be equal, and the latter be at its lowest terms, prove that $A = \mu a$, $B = \mu b$, where μ is an integer.

Proof. Suppose $A, B, a, b \in \mathbb{Z}$ such that $A/B = a/b$ and a/b is at its lowest terms. Taking $A/B = a/b$ and multiplying both sides by b and B we get $Ab = aB$. This means a divides Ab . Since a/b is at its lowest terms, a and b have no common measure save unity; that is, a is prime to b (§6). Therefore a divides A (§10, Cor. 1), which means $A = a\mu$ for some integer μ . Substituting in $Ab = aB$, $\mu ab = aB$, whence $B = \mu b$. Therefore $A = a\mu$ and $B = b\mu$ for some integer μ , completing the proof. $\square$

2. Prove that the sum or difference of two odd numbers is always even; the sum or difference of an odd and an even number always odd; the product of any number of odd numbers always odd; the quotient of one odd number by another always odd, if it be integral.

3. If a be prime to b , then—

- a) $(a + b)^m$ and $(a - b)^m$ have at most the G.C.M. 2^m ;
- b) $a^m + b^m$ and $a^m - b^m$ have at most the G.C.M. 2;
- c) $a + b$ and $a^2 + b^2 - ab$ have at most the G.C.M. 3.
4. The difference of the squares of any two odd numbers is exactly divisible by 8.
5. The sum of the squares of three consecutive odd numbers increased by 1 is a multiple of 12.
6. If each of two fractions be at its lowest terms, neither their sum nor their difference can be an integer unless the denominators be equal.
7. Resolve 45738 and 297675 into their prime factors.
8. Find the G.C.M. of 54643 and 91319, using negative remainders whenever it is of advantage to do so.
9. Prove that the L.C.M. of two integers is the quotient of their product by the G.C.M.
10. If g_1, g_2, g_3 be the G.C.M.'s, l_1, l_2, l_3 the L.C.M.'s, of b and c , c and a , a and b respectively, G the G.C.M., and L the L.C.M., of the three a, b, c , show that
- a) $L = \frac{abcG}{g_1g_2g_3}$;
- b) $\frac{L}{G} = \sqrt{\frac{l_1l_2l_3}{g_1g_2g_3}}$.
11. When x is divided by y , the quotient is u and the remainder v ; show that, when x and uy are divided by v , the remainders are the same, and the quotients differ by unity.